# Supplementary Material for

## Conditional cost–emission envelopes for decarbonizing dense urban power systems under spatial renewable scarcity and flexibility constraints

This Supplementary Material provides additional data provenance, parameter definitions, model-setting details, and extended results supporting the Xinwu District analysis. It is organized to document the building-to-node demand pipeline, renewable-potential assumptions, spatial aggregation, network abstraction, temporal representation, storage and flexibility modelling, uncertainty analysis, marginal-abatement and constraint-relaxation results, and reproducibility materials.

## S1 Demand modelling and spatial aggregation checks

The demand-side input package represents hourly electricity demand for 55,284 buildings in Xinwu District. Building-level loads are first assigned to 250 m grid cells and then aggregated into planning-node representations with 70, 100, and 150 nodes. The k100 aggregation is used as the baseline planning representation in the main model.

This section evaluates the aggregation in three steps. First, the land-use maps compare the spatial composition of k70, k100, and k150 nodes. Second, the retained aggregate-load comparison verifies that the independently materialized node-hour tables preserve the same district total across the three spatial resolutions retained in the data package. Third, the distributional checks compare intra-node load variability, node floor area, and node annual-load distributions across the three spatial resolutions. Together, these checks characterize whether the planning-node representation retains the dominant demand and land-use heterogeneity needed for district-scale capacity-expansion analysis.

These aggregation-consistency checks characterize the retained planning-node input package and the distribution of demand across k70, k100, and k150 representations. Feeder-level measured-load calibration requires restricted utility metering data and is therefore outside the retained public data package.

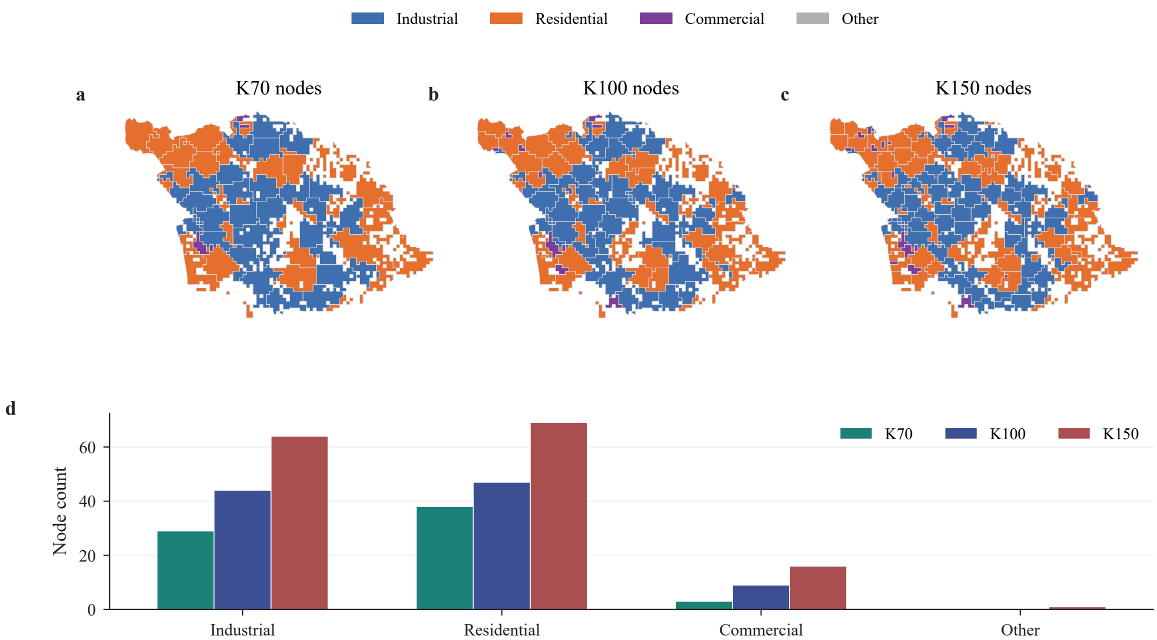


**Figure S1:** Land-use composition across k70, k100, and k150 aggregations. Maps show dominant node land use; bars compare node counts by land-use class.

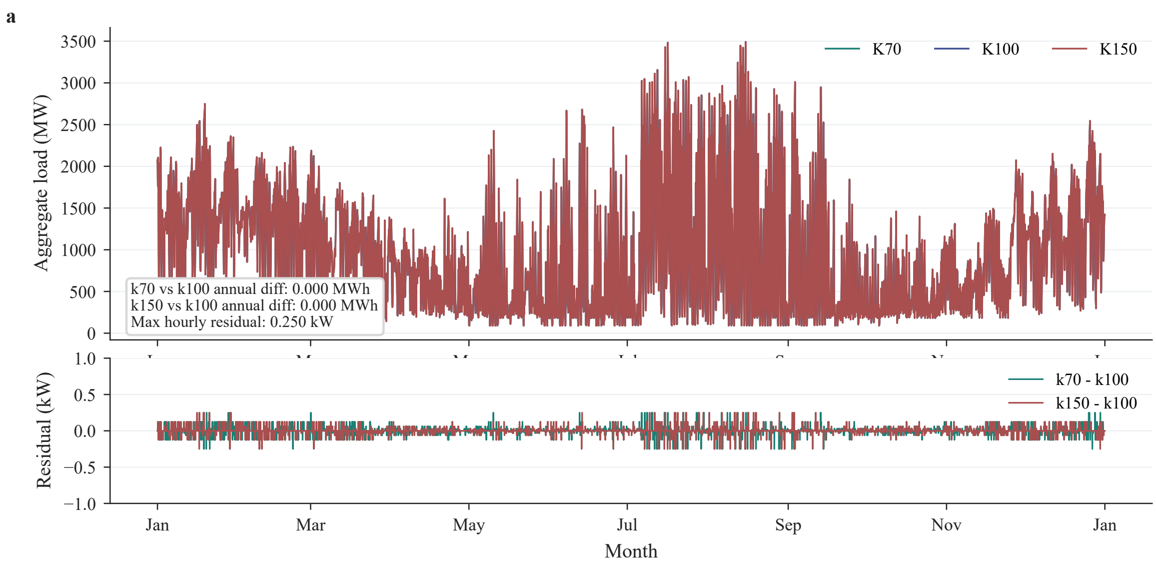


**Figure S2:** Aggregate-load consistency across the retained aggregation package. The upper trace compares the full-year aggregate hourly load across k70, k100, and k150 node-hour tables, and the lower trace shows pairwise residuals relative to k100.

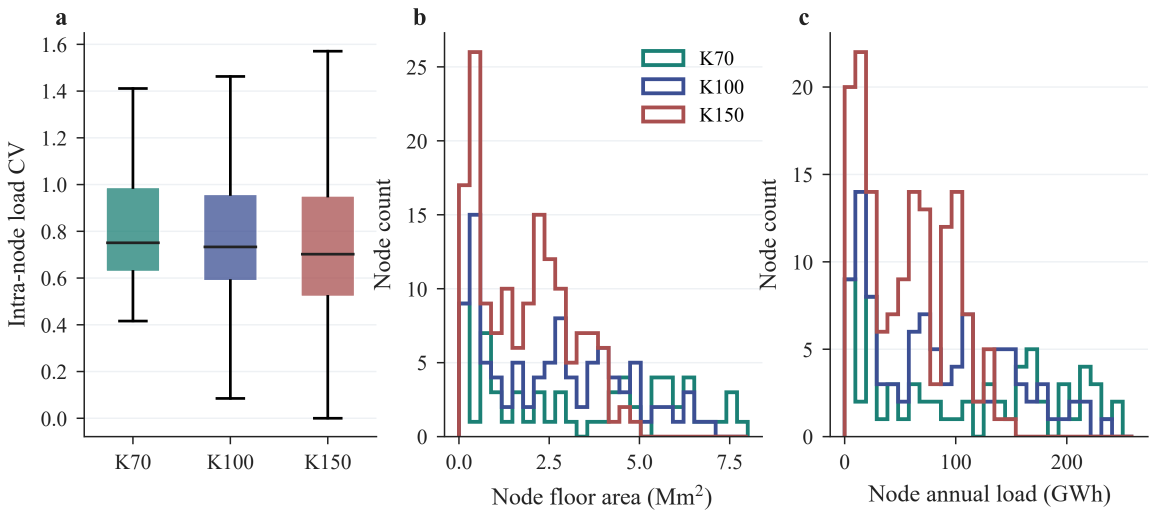


**Figure S3:** Distributional preservation across k70, k100, and k150 aggregations. Panels show intra-node load CV, node floor area, and node annual-load distributions.

Supplementary Figs. S1–S3 show land-use preservation, aggregate-load consistency, intra-node load dispersion, and the floor-area and load distributions across k70, k100, and k150. Tables S1 and S2 retain only the numerical scale of the demand package and the cross-resolution aggregation metrics.

**Table S1:** Demand package scale and aggregate consistency.

| Metric | Value | Unit |
|---|---|---|
| Building records | 55,284 | buildings |
| Populated grid cells | 2,889 | 250 m cells |
| Baseline planning nodes | 100 | nodes |
| Temporal coverage | 2017 full year | hourly |
| Demand boundary | Electricity | – |
| Annual demand | 8.234 | TWh |
| Peak load | 3.492 | GW |
| Annual residual k70–k100 | 0.000 | MWh |
| Annual residual k150–k100 | 0.000 | MWh |
| Peak residual k70–k100 | 0.000 | MW |
| Peak residual k150–k100 | 0.000 | MW |
| Maximum hourly residual | 0.25 | kW |

*Data source.* The demand package is derived from the UBEM-based demand integration workflow and the retained node-hour aggregation outputs. Source files are documented in the repository README and data-provenance files.

*Note.* The direct pre-node hourly grid-cell series is not retained in the available package. The residual rows therefore compare the retained k70 and k150 node-hour totals against the k100 baseline.

**Table S2:** Spatial aggregation metrics across k70, k100, and k150.

| Metric | k70 | k100 | k150 |
|---|---|---|---|
| Planning nodes | 70 | 100 | 150 |
| Assigned grid cells | 2,889 | 2,889 | 2,889 |
| Mapped buildings | 55,284 | 55,284 | 55,284 |
| Total floor area ($Mm^2$) | 267.16 | 267.16 | 267.16 |
| Annual demand (TWh) | 8.234 | 8.234 | 8.234 |
| Mean node load (GWh) | 117.63 | 82.34 | 54.89 |
| Median node load (GWh) | 119.13 | 72.33 | 54.61 |
| Maximum node load (GWh) | 336.56 | 234.36 | 148.82 |
| Annual-load CV | 0.759 | 0.767 | 0.723 |
| Annual-load Gini | 0.433 | 0.435 | 0.413 |
| Mean intra-node load CV | 0.806 | 0.788 | 0.739 |
| Median intra-node load CV | 0.751 | 0.733 | 0.702 |

*Note.* Floor area is reported in million square metres and annual demand in terawatt-hours to keep the table compact. Distributional detail is shown in Supplementary Fig. S3.

# S2 Renewable-potential assumptions and plausibility checks

PV and wind inputs enter the optimization as node-level technical capacity bounds before dispatch is optimized. The PV bound is derived from usable rooftop availability, a rooftop usability ratio, PV capacity density, and hourly PV availability factors. Candidate wind potential is treated as a site-conditioned urban resource bound based on retained land-use siting assumptions and hourly wind availability. These bounds limit installed capacity in the optimization; they are not interpreted as dispatchable annual energy that can be used without temporal, grid-interface, storage, and curtailment constraints.

Tables S3 and S4 report the capacity-density, capacity-factor, cost, and lifetime assumptions used for PV and wind. Table S5 provides plausibility checks by comparing technical potential with annual district demand and existing grid-interface capacity. These checks document the physical scale of the renewable-potential assumptions that underlie the Xinwu frontier.

**Table S3:** PV technical-potential assumptions.

| Parameter | Value | Unit |
|---|---|---|
| Usable rooftop ratio | 0.70 | – |
| PV capacity density | 0.200 | kW $m^{-2}$ |
| Total PV technical potential | 5.178 | GW |
| Mean PV capacity factor | 0.1663 | – |
| Annual PV generation potential | 7.545 | TWh |
| Node-level application | Capacity bound | – |

**Table S4:** Wind technical-potential assumptions.

| Parameter | Value | Unit |
|---|---|---|
| Total wind technical potential | 0.925 | GW |
| Mean wind capacity factor | 0.2323 | – |
| Annual wind generation potential | 1.882 | TWh |
| Wind base capex | 10,000 | CNY $kW^{-1}$ |
| Wind lifetime | 25 | yr |
| Node-level application | Capacity bound | – |

**Table S5:** Renewable-potential plausibility metrics.

| Metric | Value | Unit |
|---|---|---|
| Implied PV panel area | 25.891 | $km^2$ |
| Implied rooftop area | 36.986 | $km^2$ |
| Annual district demand | 8.234 | TWh |
| PV adequacy ratio | 0.916 | – |
| Wind adequacy ratio | 0.229 | – |
| Combined renewable adequacy ratio | 1.145 | – |
| Existing grid-interface capacity | 3.851 | GW |
| Grid-interface adequacy ratio | 1.103 | – |

Tables S3–S5 report the numerical assumptions and plausibility metrics. The data provenance for roof-area inputs, PV capacity factors, wind profiles, and technology costs is documented in the repository README and data-provenance files.

# S3 Spatial aggregation and network-abstraction inputs

This section documents the planning-node and network-abstraction inputs used to evaluate spatial sensitivity. The purpose is to separate three concepts that can otherwise be conflated: optimization feasibility under a spatial resolution, chronological-replay screening under the retained stress-period package, and planning-scale internal transfer availability under the network abstraction. The k70, k100, and k150 cases test how node resolution affects the represented demand and resource heterogeneity, while the node-interface, copperplate, and spatial-transfer cases bracket internal transfer availability.

Tables S6–S9 retain the numerical summaries used by the main-text spatial and network results. Supplementary Fig. S4 summarizes the cap-resolution screening matrix and the centroid-link distance distribution used to construct the spatial-transfer proxy.

**Table S6:** Spatial-resolution metrics used in the optimization sensitivity.

| Metric | k70 | k100 | k150 |
|---|---|---|---|
| Nodes | 70 | 100 | 150 |
| Annual demand (TWh) | 8.234 | 8.234 | 8.234 |
| Annual-load CV | 0.759 | 0.767 | 0.723 |
| Annual-load Gini | 0.433 | 0.435 | 0.413 |
| Mean intra-node load CV | 0.806 | 0.788 | 0.739 |

**Table S7:** Cap-resolution feasibility screen.

| Resolution | 1.00 | 0.40 | 0.25 | 0.15 | 0.14 |
|---|---|---|---|---|---|
| k70 | F | F | LS | LS | CR |
| k100 | F | F | F | F | CR |
| k150 | F | F | LS | LS | CR |

*Note.* F denotes feasible. LS denotes screened by load shedding. CR denotes screened by chronological-replay availability. NE denotes not evaluated in the selected replay configuration. The CR flag identifies the chronological-replay screening boundary for the retained stress-period package. The flag records the stress-period availability condition and leaves feeder-level network infeasibility and full-year chronological feasibility outside this screening table.

**Table S8:** Network-abstraction cases at the strict cap.

| Case | Internal transfer representation | Cost (bn CNY $yr^{-1}$) | BESS (GWh) |
|---|---|---|---|
| Node-interface | Node-specific grid-interface balance | 10.723 | 31.27 |
| Copperplate | Unrestricted internal balancing | 6.633 | 11.62 |
| Spatial-transfer proxy | $k$-nearest centroid links, 30 MW | 7.532 | 16.78 |

**Table S9:** Spatial-transfer proxy summary.

| Metric | Value |
|---|---|
| Raw edge rows | 1,194 |
| Unique undirected edges | 199 |
| Nearest-neighbour setting | 3 |
| Uniform transfer bound | 30 MW |
| Minimum centroid distance | 1.000 |
| Maximum centroid distance | 3.000 |

*Note.* The full edge-list schema and source-file path are documented in the repository README and data-provenance files. The spatial-transfer proxy is constructed from planning-node centroids. For each node, the three nearest neighbouring centroids are selected, and directed neighbour records are symmetrized into a set of unique undirected links. The retained table reports 1,194 raw edge rows and 199 unique undirected links. Each unique link uses a uniform 30 MW transfer bound. The proxy tests sensitivity to limited internal spatial balancing at the planning scale. AC distribution-network assessment requires feeder topology, electrical parameters, voltage constraints, protection settings, and reliability criteria.

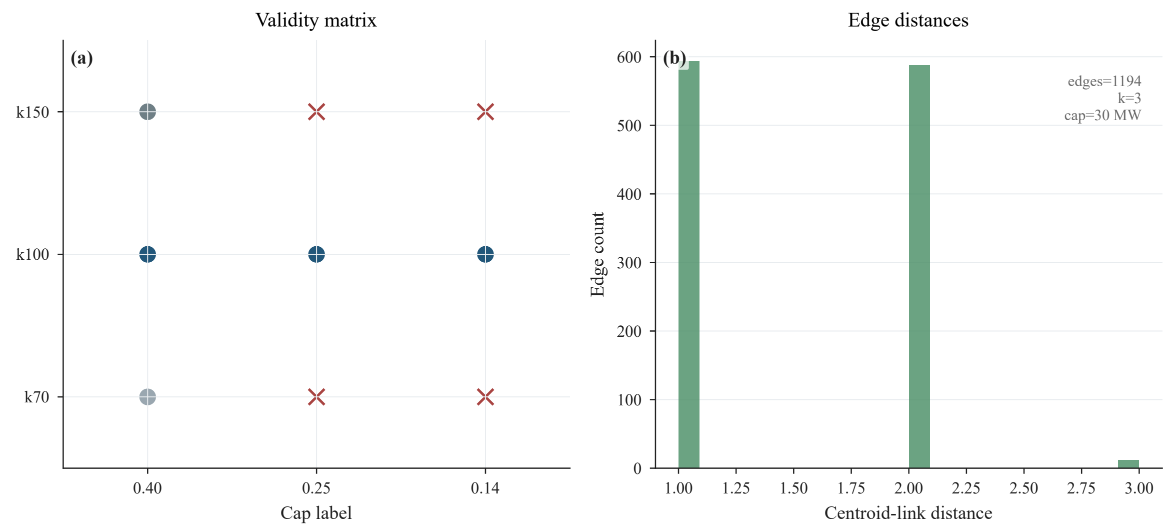


**Figure S4:** Spatial-resolution feasibility and spatial-transfer proxy summary. The figure summarizes cap-resolution feasibility outcomes and the centroid-based transfer-link construction used in the spatial-transfer proxy.

## S4 Temporal representation and chronological replay

The baseline formulation uses representative periods for capacity expansion and selected 168 h chronological replays for operating-period sensitivity. Table S10 reports the compact temporal configuration, while Tables S11 and S12 report the retained sensitivity and replay outcomes. Supplementary Fig. S5 provides the corresponding visual summary.

**Table S10:** Representative-period configuration.

| Setting | Value |
|---|---|
| Baseline temporal mode | Representative days |
| Stored baseline days | 8 |
| Weight normalization | 8,760 h |
| Sensitivity cases | 7, 14, 21, 30 days |
| Selection feature | Daily district-load profile |

*Note.* The stored representative days are 2017-01-05, 2017-01-22, 2017-04-02, 2017-05-23, 2017-07-27, 2017-07-30, 2017-10-19, and 2017-11-11 with annual weights 63, 27, 26, 66, 66, 26, 65, and 26.

**Table S11:** Representative-period sensitivity.

| Days | Cap | Emissions (Mt $CO_2$) | Cost (bn CNY $yr^{-1}$) | BESS (GWh) |
|---|---|---|---|---|
| 7 | 0.14 | 0.688 | 5.485 | 10.66 |
| 8 | 0.14 | 0.676 | 10.723 | 31.27 |
| 14 | 0.14 | 0.688 | 14.534 | 53.62 |
| 21 | 0.14 | 0.688 | 11.456 | 39.24 |
| 30 | 0.14 | 0.688 | 12.678 | 43.93 |
| 7 | 0.25 | 1.228 | 4.591 | 3.55 |
| 8 | 0.25 | 1.207 | 6.596 | 11.69 |
| 14 | 0.25 | 1.229 | 8.529 | 20.42 |
| 21 | 0.25 | 1.229 | 6.601 | 14.11 |
| 30 | 0.25 | 1.229 | 6.797 | 16.04 |

**Table S12:** Selected chronological replay windows and fixed-capacity outcomes.

| Cap | Window | Renew./load | Prorated cap (kt $CO_2$) | Emissions (kt $CO_2$) | Load shedding (GWh) | Peak import (GW) |
|---|---|---|---|---|---|---|
| 0.14 | High PV | 1.095 | 17.68 | 17.68 | 1.48 | 1.098 |
| 0.14 | Peak load | 0.640 | 23.98 | 23.98 | 83.66 | 1.556 |
| 0.14 | Renewable scarcity | 0.314 | 19.24 | 19.24 | 133.14 | 1.887 |
| 0.25 | High PV | 1.095 | 31.57 | 31.57 | 36.16 | 1.960 |
| 0.25 | Peak load | 0.640 | 42.83 | 42.83 | 93.54 | 2.093 |
| 0.25 | Renewable scarcity | 0.314 | 34.36 | 34.36 | 122.58 | 1.916 |

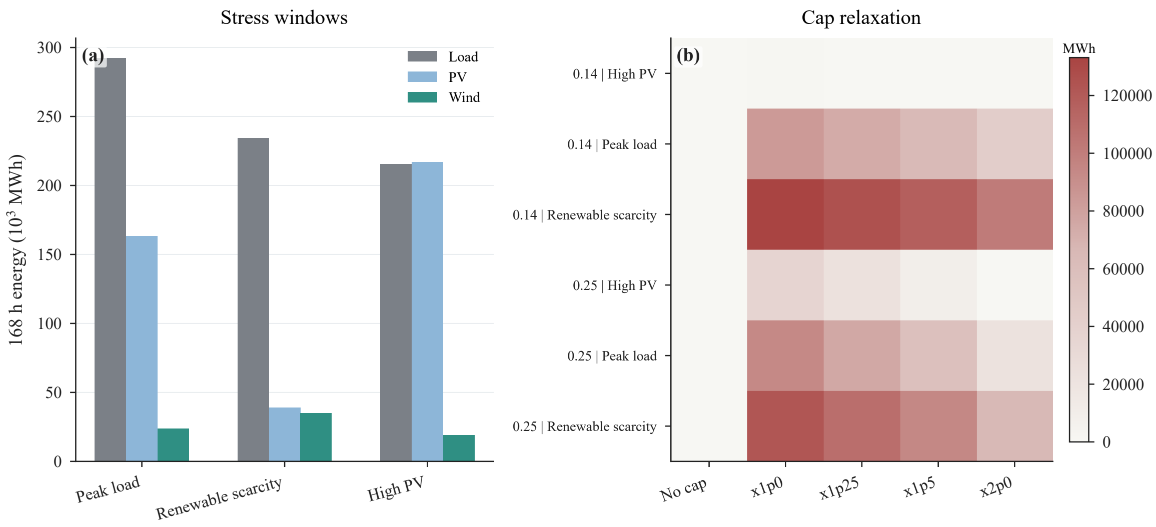


**Figure S5:** Selected chronological replay windows and cap-relaxation results. Fixed representative-period capacities are evaluated over 168 h peak-load, renewable-scarcity, and high-PV periods to characterize temporal sensitivity under concentrated operating conditions.

# S5 Storage lifecycle and flexibility assumptions

Storage is parameterized through power and energy capex, duration, efficiency, lifetime, throughput degradation, and calendar-ageing charges. Table S13 reports the compact assumption set. Tables S14 and S15 report the retained storage and flexibility outcomes, while Supplementary Fig. S6 shows the full calendar-ageing sensitivity.

**Table S13:** Storage lifecycle and duration assumptions.

| Parameter | Li-ion | LDES | Unit |
|---|---|---|---|
| Power capex | 700 | 1,600 | CNY kW$^{-1}$ |
| Energy capex | 1,200 | 240 | CNY kWh$^{-1}$ |
| Charge efficiency | 0.95 | 0.82 | – |
| Discharge efficiency | 0.95 | 0.82 | – |
| Lifetime | 12 | 25 | yr |
| Baseline duration | 4 | 24 | h |
| Extended Li-ion duration | 8 | – | h |
| Calendar-ageing charges | 0, 6, 12, 24 | – | CNY kWh$^{-1}$ yr$^{-1}$ |
| Cycling-throughput charge | 0.05 | – | CNY kWh$^{-1}$ |

**Table S14:** Storage sensitivity outcomes.

| Case | Cap | Cost (bn CNY yr$^{-1}$) | BESS (GWh) | LDES (GWh) | EFC | Calendar (CNY kWh$^{-1}$ yr$^{-1}$) |
|---|---|---|---|---|---|---|
| Li-ion 4 h legacy | 0.14 | 10.011 | 31.37 | 0.00 | 132.9 | 0 |
| Li-ion 4 h throughput | 0.14 | 10.347 | 31.29 | 0.00 | 131.6 | 0 |
| Li-ion 4 h cal. low | 0.14 | 10.535 | 31.29 | 0.00 | 131.6 | 6 |
| Li-ion 4 h cal. mid | 0.14 | 10.723 | 31.27 | 0.00 | 131.7 | 12 |
| Li-ion 4 h cal. high | 0.14 | 11.098 | 31.27 | 0.00 | 131.7 | 24 |
| Li-ion 8 h cal. mid | 0.14 | 10.357 | 31.43 | 0.00 | 131.4 | 12 |
| Li-ion 8 h + LDES | 0.14 | 8.080 | 18.39 | 19.48 | 158.2 | 12 |
| Li-ion 4 h legacy | 0.25 | 6.156 | 12.06 | 0.00 | 320.2 | 0 |
| Li-ion 4 h throughput | 0.25 | 6.455 | 11.70 | 0.00 | 279.1 | 0 |
| Li-ion 4 h cal. low | 0.25 | 6.526 | 11.70 | 0.00 | 279.1 | 6 |
| Li-ion 4 h cal. mid | 0.25 | 6.596 | 11.69 | 0.00 | 279.3 | 12 |
| Li-ion 4 h cal. high | 0.25 | 6.736 | 11.69 | 0.00 | 279.3 | 24 |
| Li-ion 8 h cal. mid | 0.25 | 6.540 | 11.79 | 0.00 | 277.3 | 12 |
| Li-ion 8 h + LDES | 0.25 | 4.574 | 0.00 | 25.20 | 0.0 | 12 |

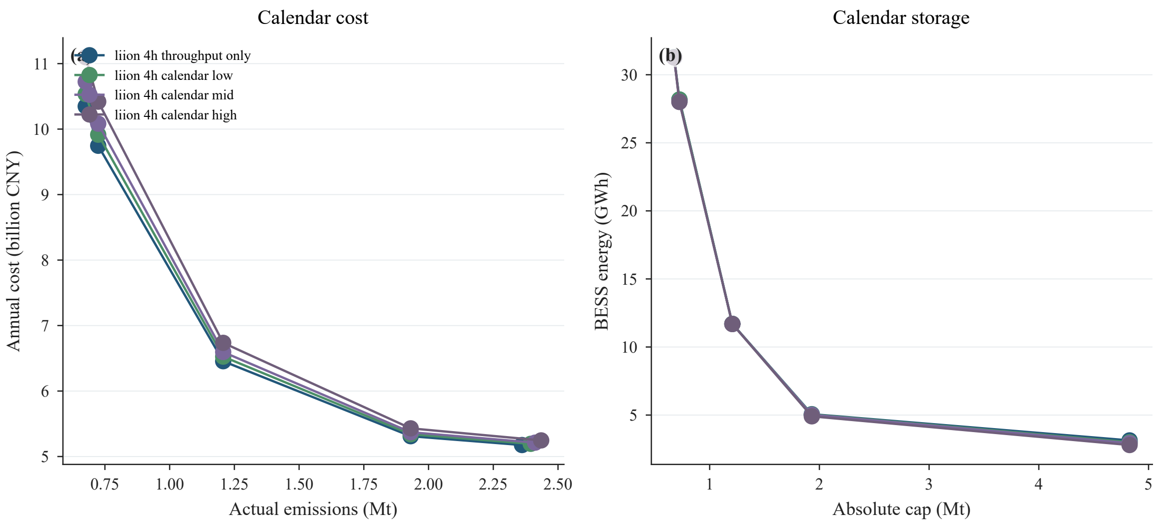


**Figure S6:** Extended storage calendar-ageing sensitivity. The figure compares storage lifecycle cases across emission-cap levels and complements the main-text storage frontier by listing the full calendar-ageing sensitivity set.

**Table S15:** Flexibility outcomes.

| Case | Cap | Cost (bn CNY $yr^{-1}$) | BESS (GWh) | DR shift (GWh) | Thermal shift (GWh) | DR peak (MW) | Thermal peak (MW) |
|---|---|---|---|---|---|---|---|
| Baseline | 0.14 | 10.723 | 31.27 | 0.00 | 0.00 | 0.0 | 0.0 |
| DR only | 0.14 | 10.687 | 31.15 | 76.09 | 0.00 | 131.7 | 0.0 |
| Thermal only | 0.14 | 10.686 | 31.15 | 0.00 | 76.09 | 0.0 | 131.7 |
| DR + thermal | 0.14 | 10.654 | 31.05 | 67.73 | 74.90 | 133.0 | 135.4 |
| Baseline | 0.25 | 6.596 | 11.69 | 0.00 | 0.00 | 0.0 | 0.0 |
| DR only | 0.25 | 6.563 | 11.56 | 88.14 | 0.00 | 142.4 | 0.0 |
| Thermal only | 0.25 | 6.562 | 11.56 | 0.00 | 88.15 | 0.0 | 142.4 |
| DR + thermal | 0.25 | 6.532 | 11.45 | 77.84 | 86.84 | 142.4 | 142.4 |

# S6 Extended uncertainty analysis

The uncertainty experiment uses Latin hypercube sampling with seed 42 and 150 requested samples. PRCC values are computed for the retained valid rows at cap labels 0.25 and 0.14, with 60 retained valid rows at cap 0.25 and 31 retained valid rows at cap 0.14. Table S16 reports the sampled ranges, while Table S17 reports the retained signed PRCC coefficients. The sampling design treats uncertain inputs as independent because district-specific empirical joint distributions linking demand scale, renewable-resource bias, technology costs, tariff levels, grid-interface capacity, flexibility uptake, and grid-carbon factors are not available in the retained data package. The PRCC rankings are therefore interpreted as conditional monotonic sensitivity rankings under independent LHS sampling.

**Table S16:** Uncertain input ranges.

| Parameter | Range | Unit |
|---|---|---|
| Demand scale | 0.901–1.099 | multiplier |
| PV potential scale | 0.802–1.198 | multiplier |
| PV capacity-factor scale | 0.900–1.099 | multiplier |
| Wind potential scale | 0.701–1.300 | multiplier |
| Wind capacity-factor scale | 0.801–1.198 | multiplier |
| Grid-interface capacity scale | 0.802–1.197 | multiplier |
| Grid carbon-factor scale | 0.803–1.200 | multiplier |
| PV capex | 4,001.739–6,495.416 | CNY $kW^{-1}$ |
| Wind capex | 9,000.217–11,993.936 | CNY $kW^{-1}$ |
| BESS power capex | 501.746–998.499 | CNY $kW^{-1}$ |
| BESS energy capex | 1,001.138–1,497.222 | CNY $kWh^{-1}$ |
| BESS degradation cost | 0.001–0.120 | CNY $kWh^{-1}$ |
| Time-of-use tariff scale | 0.851–1.199 | multiplier |
| Demand-response fraction | 0.001–0.150 | share |

**Table S17:** Top-ranked signed PRCC coefficients.

| Cap | Outcome | Parameter | PRCC | *n* |
|---|---|---|---|---|
| 0.14 | Emissions | BESS energy capex | -0.626 | 31 |
| 0.14 | Emissions | PV potential scale | 0.594 | 31 |
| 0.14 | Emissions | Grid-interface capacity scale | 0.553 | 31 |
| 0.14 | Emissions | Demand scale | -0.416 | 31 |
| 0.14 | Emissions | Demand-response fraction | -0.342 | 31 |
| 0.14 | Emissions | Wind capex | -0.245 | 31 |
| 0.14 | Emissions | Time-of-use tariff scale | 0.223 | 31 |
| 0.14 | Emissions | BESS power capex | -0.202 | 31 |
| 0.14 | Emissions | Wind capacity-factor scale | -0.168 | 31 |
| 0.14 | Emissions | Grid carbon-factor scale | -0.142 | 31 |
| 0.14 | BESS | Grid carbon-factor scale | 0.847 | 31 |
| 0.14 | BESS | Demand scale | 0.843 | 31 |
| 0.14 | BESS | PV potential scale | -0.825 | 31 |
| 0.14 | BESS | PV capacity-factor scale | -0.641 | 31 |
| 0.14 | BESS | Wind capacity-factor scale | -0.436 | 31 |
| 0.14 | BESS | Grid-interface capacity scale | -0.332 | 31 |
| 0.14 | BESS | BESS power capex | 0.246 | 31 |
| 0.14 | BESS | Demand-response fraction | 0.213 | 31 |
| 0.14 | BESS | BESS energy capex | -0.206 | 31 |
| 0.14 | BESS | Time-of-use tariff scale | 0.130 | 31 |
| 0.14 | Grid import | Grid carbon-factor scale | -1.000 | 31 |
| 0.14 | System cost | Demand scale | 0.879 | 31 |
| 0.14 | System cost | Grid carbon-factor scale | 0.839 | 31 |
| 0.14 | System cost | PV capacity-factor scale | -0.829 | 31 |
| 0.14 | System cost | PV potential scale | -0.818 | 31 |
| 0.14 | System cost | BESS energy capex | 0.723 | 31 |
| 0.14 | System cost | PV capex | 0.603 | 31 |
| 0.14 | System cost | BESS degradation cost | 0.484 | 31 |
| 0.14 | System cost | Grid-interface capacity scale | -0.403 | 31 |
| 0.14 | System cost | Demand-response fraction | 0.388 | 31 |
| 0.14 | System cost | BESS power capex | 0.388 | 31 |
| 0.25 | Emissions | Grid-interface capacity scale | -0.259 | 60 |
| 0.25 | Emissions | PV capacity-factor scale | -0.247 | 60 |
| 0.25 | Emissions | Wind capacity-factor scale | 0.239 | 60 |
| 0.25 | Emissions | Wind potential scale | 0.174 | 60 |
| 0.25 | Emissions | BESS energy capex | 0.162 | 60 |
| 0.25 | Emissions | Grid carbon-factor scale | -0.112 | 60 |
| 0.25 | Emissions | PV capex | -0.074 | 60 |
| 0.25 | Emissions | Time-of-use tariff scale | 0.068 | 60 |
| 0.25 | Emissions | Demand scale | 0.044 | 60 |
| 0.25 | Emissions | BESS power capex | -0.041 | 60 |
| 0.25 | BESS | Demand scale | 0.959 | 60 |
| 0.25 | BESS | Grid carbon-factor scale | 0.943 | 60 |

| Cap | Outcome | Parameter | PRCC | $n$ |
|---|---|---|---|---|
| 0.25 | BESS | PV potential scale | -0.875 | 60 |
| 0.25 | BESS | PV capacity-factor scale | -0.621 | 60 |
| 0.25 | BESS | BESS power capex | -0.197 | 60 |
| 0.25 | BESS | BESS degradation cost | -0.161 | 60 |
| 0.25 | BESS | Demand-response fraction | 0.157 | 60 |
| 0.25 | BESS | Time-of-use tariff scale | 0.151 | 60 |
| 0.25 | BESS | Grid-interface capacity scale | 0.097 | 60 |
| 0.25 | BESS | Wind potential scale | -0.077 | 60 |
| 0.25 | Grid import | Grid carbon-factor scale | -1.000 | 60 |
| 0.25 | System cost | Demand scale | 0.965 | 60 |
| 0.25 | System cost | Grid carbon-factor scale | 0.888 | 60 |
| 0.25 | System cost | PV potential scale | -0.825 | 60 |
| 0.25 | System cost | PV capex | 0.778 | 60 |
| 0.25 | System cost | PV capacity-factor scale | -0.757 | 60 |
| 0.25 | System cost | BESS energy capex | 0.750 | 60 |
| 0.25 | System cost | BESS degradation cost | 0.562 | 60 |
| 0.25 | System cost | Wind capacity-factor scale | -0.424 | 60 |
| 0.25 | System cost | Time-of-use tariff scale | 0.403 | 60 |
| 0.25 | System cost | BESS power capex | 0.301 | 60 |

*Note.* The complete machine-readable PRCC table is documented in the repository README and data-provenance files.

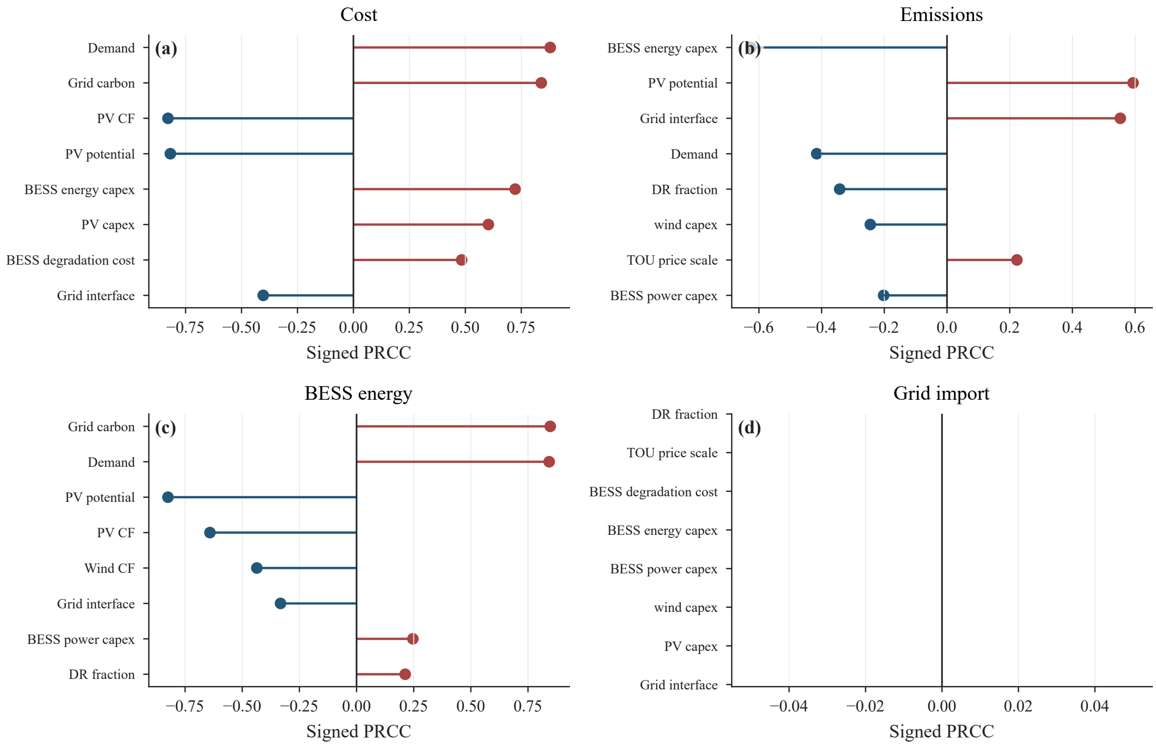


**Figure S7:** Extended PRCC rankings by outcome and emission-cap level. Signed coefficients are retained to indicate whether each uncertain input increases or decreases the corresponding cost, emissions, storage, or import metric.

# S7 Marginal-abatement and limiting-constraint results

This section provides the numerical detail behind the marginal-abatement and bottleneck diagnostics in the main text. The finite-difference MAC values describe interval-average cost changes between adjacent frontier points, while the dual-based MAC values describe local shadow-price information for the $CO_2$ cap constraint in the retained linear formulation. The limiting-constraint counts identify which constraint families bind most frequently at each cap level, and the 1% relaxation tests quantify how annualized system cost changes when selected resource, interface, or storage-duration bounds are relaxed. Together, these results support the interpretation of the cost–emission envelope as a diagnostic object instead of only a descriptive frontier.

**Table S18:** MAC comparison by cap.

| Cap | Finite-difference MAC (CNY $tCO_2^{-1}$) | Dual-based MAC (CNY $tCO_2^{-1}$) | Difference (CNY $tCO_2^{-1}$) |
|---|---|---|---|
| 0.40 | 330.6 | 850.3 | 519.7 |
| 0.25 | 1,691.6 | 2,773.4 | 1,081.8 |
| 0.15 | 7,224.0 | 12,785.2 | 5,561.2 |
| 0.14 | 13,256.6 | 13,304.4 | 47.8 |

**Table S19:** Binding counts by cap.

| Cap | Constraint family | Binding count |
|---|---|---|
| 0.25 | BESS charge power | 1,366 |
| 0.25 | BESS discharge power | 12 |
| 0.25 | BESS duration | 100 |
| 0.25 | BESS energy bound | 2,020 |
| 0.25 | $CO_2$ cap | 1 |
| 0.25 | Grid interface | 284 |
| 0.14 | BESS charge power | 2 |
| 0.14 | BESS discharge power | 0 |
| 0.14 | BESS duration | 100 |
| 0.14 | BESS energy bound | 802 |
| 0.14 | $CO_2$ cap | 1 |
| 0.14 | Grid interface | 320 |

**Table S20:** One-percent relaxation results.

| Cap | Relaxed family | Cost relief (bn CNY $yr^{-1}$) | Top nodes |
|---|---|---|---|
| 0.14 | PV potential | 0.0194 | k100_0084, k100_0069, k100_0029 |
| 0.14 | Battery duration | 0.0075 | GLOBAL |
| 0.14 | Wind potential | 0.0031 | k100_0084, k100_0069, k100_0049 |
| 0.14 | Grid capacity | 0.0000 | k100_0034, k100_0011, k100_0025 |
| 0.25 | PV potential | 0.0039 | k100_0084, k100_0069, k100_0029 |
| 0.25 | Battery duration | 0.0028 | GLOBAL |
| 0.25 | Wind potential | 0.0006 | k100_0084, k100_0069, k100_0049 |
| 0.25 | Grid capacity | -0.0000 | k100_0034, k100_0011, k100_0033 |

# S8 Solver settings, feasibility screening, and reproducibility materials

This section retains the repository-facing material needed for reconstruction and auditing. Tables S21–S22 keep only the compact numerical settings and screening thresholds needed to interpret the public package. Repository locations, source categories, and data-provenance details are documented in the repository README and data-provenance files.

**Table S21:** Optimization solver and numerical settings.

| Setting | Value |
|---|---|
| Solver | scipy-linprog-highs |
| Status | optimal |
| Representative runtime | 29.91 s |
| Objective at cap 0.14 | 10.723 bn CNY $yr^{-1}$ |
| Stored representative days | 8 |
| Load-shedding penalty | 100,000 CNY $kWh^{-1}$ |
| Discount rate | 0.06 |
| Planning horizon | 20 yr |
| PV capex / lifetime | 5,000 CNY $kW^{-1}$ / 25 yr |
| Wind capex / lifetime | 10,000 CNY $kW^{-1}$ / 25 yr |

**Table S22:** Feasibility-screening thresholds.

| Criterion | Threshold | Purpose |
|---|---|---|
| Load shedding | $\leq 10^{-6}$ MWh | Accepted row |
| $CO_2$ violation | $\leq 10^{-4}$ ton | Accepted row |
| Cost consistency | $10^{-4}$ CNY | Reconstructed cost |
| Objective consistency | max($10^{-4}$ CNY, $10^{-10}$ relative) | Solver check |
| Zero-storage cost | flagged if inconsistent | Storage check |